

The Verified-Candidate Loop for Budgeted Large Language Model-Assisted High-Level Synthesis

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Abstract—High-level synthesis (HLS) agents must repair and optimize unfamiliar kernels under limited model and tool budgets. The Verified-Candidate Loop (VCL) retains a verified fallback, promotes a candidate only after seven tool gates, and ranks valid candidates with a bounded composite score. Deterministic diagnostic extraction and public-evidence retrieval constrain later model proposals. We evaluate three deployed agent–endpoint configurations on a frozen, balanced 150-task suite. A deterministic retry rule replaces only records already classified as LLM-API infrastructure errors. Across the final records, 424 endpoint-task outputs reach public completion; all 424 pass the evaluator’s finite hidden gate suite. DeepSeek V4 Pro has the largest raw completion point estimate (143/150) and lowest observed selected-record token total (2.128M), whereas Qwen3.5-122B-A10B has the largest API-conditioned completion point estimate (140/145) and paired QoR means (77.77, 78.93) on 24 common tasks. Single campaigns and source-provenance differences make these descriptive configuration results, not causal model comparisons.

Index Terms—high-level synthesis, LLM agent, FPGA, retrieval-augmented generation, quality of results

I. GATE EVIDENCE, NOT PROMPTS

LLM agents can edit unfamiliar HLS kernels, but prompts alone do not establish correctness or hardware quality. Tool reports provide reproducible, but test-bounded, evidence. A metered agent therefore needs a promotion rule that preserves a verified fallback whenever a required validation gate fails [1].

Prior HLS agents automate code transformation, directive search, and feedback-driven repair [2]–[4]. Tool-using language agents alternate reasoning and actions [5]. Our design assigns candidate promotion to deterministic HLS evidence under explicit model and tool budgets.

The Verified-Candidate Loop (VCL) presents three elements. **(1) Validation contract.** A fail-closed workflow preserves a verified fallback across Interface, CSIM, SYNTH, Frequency, Capacity, required CoSIM, and Metric-completeness gates. **(2) Composite promotion utility.** The bounded Q_{HW} combines measured performance and resource ratios with source-change and repair evidence; despite its implementation name, it is not a pure hardware QoR metric. **(3) Separated evaluation views.** Raw completion, API-conditioned completion, finite hidden-gate pass, and paired QoR are reported with their own denominators (Section IV).

II. TOOL EVIDENCE CONTROLS THE AGENT WORKFLOW

The task interface exposes an editable kernel, public tests, target part, and budgets while withholding hidden tests and frozen references. The controller records every candidate, gate, metric, token, and credit in one run state.

The VCL runs seven gates in order: Interface, CSIM, SYNTH, Frequency, Capacity, required CoSIM, and Metric completeness. Frequency enforces 100 MHz, Capacity enforces U55C limits, and CoSIM applies only when required. Define validity $V(c) = \prod_i G_i$ over applicable gates $G_i \in \{0, 1\}$; an inapplicable gate contributes 1. Let b be the verified fallback. The promotion rule is $P(c) = V(c) \wedge [Q_{\text{HW}}(c) > Q_{\text{HW}}(b)]$.

Local failure classification and retrieval require no additional model call; generating a subsequent candidate still does. **Failure Reflection** parses interface, synthesis, and co-simulation diagnostics plus the source diff into a compact constraint for the next proposal. **QoR-RAG** retrieves compatible public rules and measured success/failure cases by task metadata and bottleneck, then adds them to the optimization context. Thus interface mismatches produce signature/pragma constraints, synthesis failures produce timing/resource/II constraints, and co-simulation deadlocks produce handshake/port-width constraints. Duplicate candidates and repeated failures stop retries.

Figure 1 summarises the loop. The controller records every candidate and gate outcome for audit, while a failed or incomplete candidate never replaces the last verified fallback.

III. A COMPOSITE SCORE REJECTS COSTLY GAINS

Q_{HW} is the implementation’s bounded promotion utility, not a hardware-only objective. For anchor a and candidate c , let $r_L = (p_a L_a)/(p_c L_c)$ use cycle latency L and effective clock period p . When initiation interval applies, $r_P = r_L^{0.85} (II_a/II_c)^{0.15}$; otherwise $r_P = r_L$. Define the capacity-normalized resource footprint $A(x) = \sum_r x_r/K_r$ over LUT, FF, DSP, BRAM, and URAM, and $r_A = A(a)/A(c)$. Let $D \in \{0, 1\}$ mark a source change and $R \in \{0, 1\}$ mark a valid candidate that repairs an invalid starter. The composite ratio is $h = 0.99^D 2^{R r_P^{0.55} r_A^{0.45}}$.

$Q_{\text{HW}} = 1 - 1/(1 + h)^2$ maps $h=1$ to 0.75, approaches 1 for large gains, and approaches 0 for regressions. Budget efficiency $E = \max(0.80, 1 - 0.10u_{\text{credit}} - 0.10u_{\text{time}})$ uses consumed fractions clipped to $[0, 1]$. The final score is

$$S = 100 V(c) Q_{\text{HW}} E. \quad (1)$$

The 0.99 and 2 factors encode anti-no-op and repair policies; the 0.85/0.15 and 0.55/0.45 exponents are fixed heuristic weights. The same scoring module and constants apply to every campaign, but we make no sensitivity claim; reported QoR comparisons are conditional on this utility. Promotion occurs only after $V(c)=1$, and a failed gate yields $S=0$.

IV. SEPARATED EVALUATION MEASURES

Protocol. Each configuration first processes the same frozen, balanced 150-task Track-A v4 suite [6]. A deterministic merge

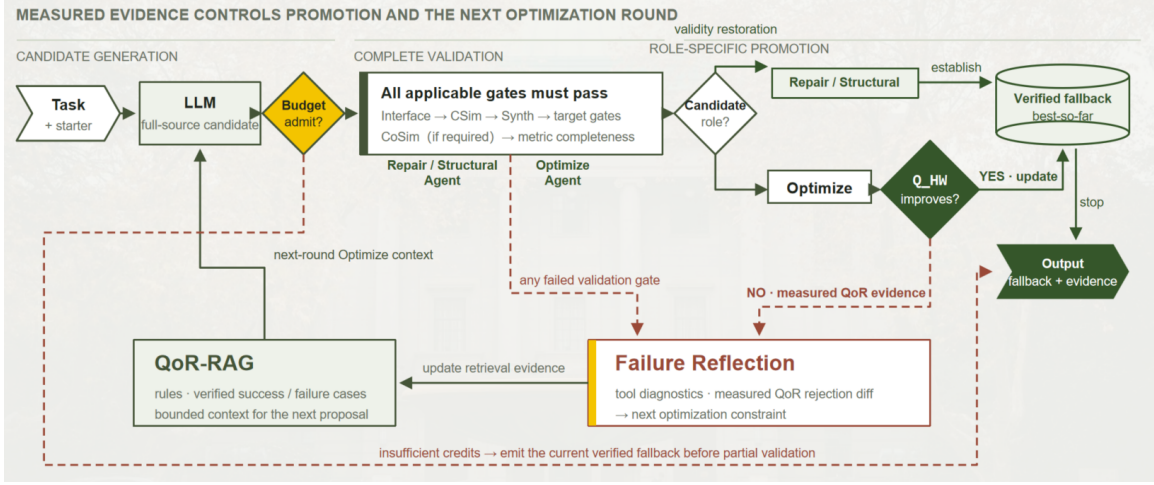


Fig. 1. **The Verified-Candidate Loop.** A candidate replaces the verified fallback only after every applicable gate passes and the composite Q_{HW} improves.

rule permits a retry only when the currently selected record is classified as an LLM-API infrastructure error; the chronological retry replaces it regardless of retry outcome. Validation failures and 900-second task timeouts are ineligible. First-pass public completions are 143, 129, and 131; after 0, 17, and 19 eligible retry records, the final counts are 143, 140, and 141.

Public completion retains all 150 tasks. API-conditioned completion excludes only unresolved LLM API-call records from its denominator and retains task timeouts and validation failures. A separate evaluator reruns each public-completed output using finite hidden CSIM, synthesis, and required CoSIM gates. The QoR columns apply Eq. (1) to the same 24 tasks.

TABLE I
FINAL DESCRIPTIVE POINT ESTIMATES. HIDDEN IS CONDITIONAL ON PUBLIC;
API-COND. REMOVES UNRESOLVED API-CALL RECORDS. QoR MEANS USE THE
PAIRED SUBSET; QWEN TOKEN TOTALS ARE LOWER BOUNDS.

Configuration	Public	API-cond.	Hidden	Tok. (M)	Proxy	Ref. QoR
DeepSeek V4 Pro	143/150	143/150	143/143	2.128	74.98	75.69
Qwen3.5-72B-A10B	140/150	140/145	140/140	≥ 3.965	77.77	78.93
Qwen3.6-72B	141/150	141/147	141/141	≥ 5.044	77.06	78.56

Results. (1) All 424/424 final outputs that pass the public gate also pass the recorded finite hidden-gate suite. The 26 endpoint-task runs without public completion are outside this denominator; the result is test-bounded evidence, not a correctness certificate.

(2) DeepSeek has the largest raw completion point estimate (95.3%); Qwen3.5 has the largest API-conditioned point estimate (96.6%). These answer different questions, so the latter is not relabelled as agent completion.

(3) Qwen3.5 has the largest paired QoR point estimates. Its reference mean is only 0.37 above Qwen3.6; one campaign per configuration provides no uncertainty or stable-ranking claim. DeepSeek has the lowest observed selected-record token total.

Configuration boundary. The recorded snapshots match in 83/86 files, including controller, prompts, gates, and scoring; differences are confined to provider/run-contract, metric-recording, and runner/record-validation plumbing. Because the

provider adapter is outside the snapshot, the rows compare complete deployed configurations rather than isolating endpoint or model effects.

V. LIMITS AND NEXT STEPS

The VCL retains a verified fallback under finite tool gates and a composite promotion utility. All 424 public-completed outputs pass the recorded hidden suite, conditional on excluding 26 runs without public completion.

Claims remain system-level: evidence comprises one Vitis 2025.2/U55C campaign per configuration, no repeats, intervals, sensitivity study, or component ablation, and QoR covers 24 common tasks under a fixed heuristic. Moreover, provider/run plumbing differs despite 83/86 matching source files; the provider adapter is unsnapshotted and the DeepSeek identity lacks immutable license evidence. Hence causal, stable-ranking, and competition-compliance claims are unwarranted.

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